

HUADE, China

WMO: 533910

Lat: 41.8956N

Lon: 114.0014E

Elev: 1484

StdP: 84.72

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.0	-23.1	-30.7	0.3	-20.9	-28.3	0.3	-19.7	11.3	-18.1	10.3	-17.3	3.7	290	0.598

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	29.2	15.7	27.5	15.1	26.0	14.6	18.9	23.6	18.0	22.8	17.1	22.1	3.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.6	15.1	20.7	16.5	14.1	20.1	15.4	13.1	19.5	60.6	23.6	57.2	22.6	54.3	22.1	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.1	8.7	7.7	-27.5	31.4	2.7	1.8	-29.5	32.7	-31.0	33.8	-32.6	34.8	-34.5	36.2
			-27.8	20.0	2.7	1.1	-29.7	20.8	-31.2	21.5	-32.7	22.1	-34.7	22.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.0	-14.3	-10.2	-2.8	5.6	12.2	17.4	20.1	18.3	12.8	4.9	-4.7	-12.2	(6)
(7)		DBStd	12.84	4.92	5.58	5.90	5.50	4.70	3.37	2.63	2.74	4.12	4.99	5.74	4.89	(7)
(8)		HDD10.0	3218	754	566	397	152	31	1	0	0	21	169	441	688	(8)
(9)		HDD18.3	5368	1012	799	654	383	196	56	12	33	169	416	691	946	(9)
(10)		CDD10.0	1026	0	0	1	19	98	222	312	258	104	11	0	0	(10)
(11)		CDD18.3	135	0	0	0	0	6	27	66	34	3	0	0	0	(11)
(12)		CDH23.3	1116	0	0	0	5	76	261	503	238	34	0	0	0	(12)
(13)		CDH26.7	232	0	0	0	1	12	57	122	37	3	0	0	0	(13)
(14)	Wind	WSAvg	3.4	3.6	3.6	3.9	4.2	4.0	3.1	2.7	2.5	2.8	3.3	3.6	3.8	(14)
(15)	Precipitation	PrecAvg	319	2	3	5	12	26	55	85	76	35	15	4	2	(15)
(16)		PrecMax	445	7	14	11	42	62	162	194	175	80	42	14	8	(16)
(17)		PrecMin	166	0	0	0	0	2	16	10	20	5	0	0	0	(17)
(18)		PrecStd	68	2	3	3	9	16	36	42	36	20	10	4	2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	0.7	6.3	15.6	23.7	28.3	30.8	31.8	29.9	26.6	20.7	11.7	2.4	(19)	
(20)		MCWB	-3.8	0.0	4.5	9.6	11.5	14.9	17.1	15.7	12.9	9.6	3.9	-2.5	(20)	
(21)		DB	-2.3	3.0	11.9	20.3	25.5	28.3	29.6	27.6	23.9	17.5	8.3	-0.9	(21)	
(22)		MCWB	-5.7	-2.0	2.9	7.7	10.9	14.0	16.8	15.9	12.6	8.3	1.9	-4.3	(22)	
(23)		DB	-4.6	0.8	9.0	17.5	23.3	26.3	27.9	25.9	22.0	15.1	5.9	-3.1	(23)	
(24)		MCWB	-7.2	-3.3	1.4	6.3	10.0	13.7	16.2	15.1	11.6	6.7	0.6	-5.8	(24)	
(25)		DB	-6.8	-1.4	6.3	15.0	21.1	24.4	26.2	24.4	20.0	12.8	3.5	-5.0	(25)	
(26)		MCWB	-8.9	-4.8	-0.1	5.2	9.1	12.9	15.8	15.1	10.8	5.5	-0.7	-7.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	-3.4	0.4	5.5	11.1	13.6	17.4	20.6	19.9	16.5	10.8	4.3	-2.3	(27)	
(28)		MCDWB	0.3	5.1	13.9	19.5	22.8	24.0	26.4	24.5	21.0	17.4	9.1	1.7	(28)	
(29)		WB	-5.5	-1.5	3.3	8.6	12.4	16.4	19.4	18.7	14.6	9.1	2.6	-4.3	(29)	
(30)		MCDWB	-2.5	2.2	10.9	18.0	21.5	23.0	24.7	23.0	20.1	15.3	7.6	-1.3	(30)	
(31)		WB	-7.3	-3.1	1.7	7.1	11.2	15.5	18.4	17.7	13.3	7.8	1.0	-5.7	(31)	
(32)		MCDWB	-4.8	0.6	8.4	15.5	20.2	22.1	23.4	21.9	18.8	13.3	5.4	-3.2	(32)	
(33)		WB	-8.9	-4.8	0.1	5.9	10.2	14.6	17.7	16.8	12.2	6.5	-0.5	-7.2	(33)	
(34)		MCDWB	-6.8	-1.5	6.0	13.7	18.4	21.1	22.8	21.1	18.0	11.6	3.4	-5.0	(34)	
(35)	Mean Daily Temperature Range	MDBR	9.4	10.4	11.0	11.4	11.6	10.8	10.0	10.0	10.6	10.3	9.6	9.0	(35)	
(36)		5% DB	12.0	12.4	13.7	14.6	14.2	13.5	12.5	12.4	12.9	13.0	11.9	11.3	(36)	
(37)		MCWBR	9.4	8.8	8.1	7.4	6.3	5.0	4.1	4.6	5.4	6.8	7.6	8.7	(37)	
(38)		5% WB	11.7	12.0	13.1	13.1	12.4	10.6	9.9	9.3	10.8	11.5	11.0	11.0	(38)	
(39)		MCWBR	9.3	8.8	8.1	7.2	6.0	4.5	4.1	4.0	5.4	6.6	7.4	8.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.269	0.315	0.353	0.409	0.452	0.493	0.465	0.452	0.409	0.356	0.299	0.278	(40)	
(41)		taud	2.306	2.178	2.104	2.007	1.937	1.907	2.047	2.055	2.104	2.184	2.275	2.278	(41)	
(42)		Ebn at Noon	882	879	884	857	828	792	811	808	815	817	828	829	(42)	
(43)		Edh at Noon	85	115	141	168	185	191	165	158	140	114	88	81	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.39	3.39	4.89	5.92	6.63	6.24	6.15	5.75	4.83	3.74	2.59	2.06	(44)	
(45)		RadStd	0.26	0.27	0.37	0.38	0.42	0.48	0.28	0.31	0.25	0.19	0.25	0.18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	-1.84	N/A	N/A	-0.50	N/A	N/A	N/A	N/A	
(47)	Regional (3 neighbors)	N/A	N/A	-1.26	N/A	-0.40	-0.59	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page